

The numerical range is a 2-spectral set

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Abstract

Let A be a complex square matrix and let $W(A)$ be its numerical range. We prove that

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|$$

for every complex polynomial p , and hence for every rational function with no pole on $W(A)$. The constant is optimal. The proof combines the classical positive double-layer functional calculus with a new positive-real completion principle. The argument first treats algebra-generating, simple-spectrum approximants. For these auxiliary objects, the double-layer map applied to the Cayley family completes the resolvent of $T = f(B)$ by a term in the algebra generated by T^* . Positivity of the resulting matrix-valued Herglotz kernel is sampled at points determined by the eigenvalues of T , and a compensating sample at the origin cancels the completion exactly. The remaining Pick-matrix inequality compares two weighted observability Gramians and, through a Stein identity, gives $\|T\| \leq 2$. Perturbation, density, and smooth convex outer approximation then pass the estimate to arbitrary matrices and their exact numerical ranges. Thus no simple-spectrum, diagonalizability, algebra-generation, or boundary-regularity condition is imposed on A .

1. Introduction

We write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}, \quad \mathbb{T} = \partial\mathbb{D},$$

use X^* for the adjoint (the conjugate transpose in finite dimensions), put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T .

For $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ be the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$(1) \quad W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

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Throughout, $\|x\|_2$ denotes the Euclidean norm of a coordinate vector, whereas $\|X\|$ denotes the induced operator norm of a matrix or bounded operator. Scalar moduli are written $|\cdot|$. The Toeplitz–Hausdorff theorem says that $W(A)$ is convex; in finite dimensions it is also compact and contains $\sigma(A)$. The numerical range therefore records both spectral and nonnormal information. The problem considered here is whether it controls the full holomorphic functional calculus with a universal, sharp constant.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$(2) \quad \|r(A)\| \leq C \max_{z \in K} |r(z)|$$

for every rational function r with no pole on K . Crouzeix proved the constant-2 assertion for 2×2 matrices and formulated the general conjecture in [14]. His subsequent dimension-free theorem gave the constant 11.08 [15]. After further structural and numerical work [17], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [22]. Their double-layer and Cauchy-transform argument was clarified in [35] and developed in several directions in [8] and [19].

The intervening literature established the conjectured constant for a variety of special classes, among them perturbed Jordan blocks, certain 3×3 nilpotent matrices, and tridiagonal 3×3 matrices with elliptic numerical range [11], [16], [10], and [25]. Numerical and extremal investigations appear in [26] and [29], and a broad account of the problem and its variants is given in [4]. More recent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [6], [34], [5], and [20]. A new proof for a class of weighted shifts appears in [21]. None of these reductions provided a dimension-free route from $1 + \sqrt{2}$ to 2.

The main result is the following sharp estimate.

THEOREM 1.1 (Main theorem). *Let $n \geq 1$, let $A \in \mathbb{M}_n(\mathbb{C})$, and let $p \in \mathbb{C}[z]$. Then*

$$(3) \quad \|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

Consequently, $W(A)$ is a 2-spectral set for A .

The statement is unconditional: A and p are arbitrary, and no diagonalizability, spectral-simplicity, algebra-generation, or boundary regularity condition is imposed.

The constant cannot be decreased. Indeed, for

$$(4) \quad N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z,$$

one has $W(N) = \overline{\mathbb{D}}$ and $\|N\| = 2$.

1.1. *Where the earlier mechanism loses sharpness.* The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [24]; see also [3] and [22]. If B has numerical range in a smooth convex domain Ω and f is bounded by one on Ω , this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$(5) \quad 2\Phi(f) = f(B) + g_f(B)^*.$$

Thus positivity directly controls a symmetrized functional calculus, not $f(B)$ by itself. The Crouzeix–Palencia argument couples the two terms in (5) and gives the sharp conclusion of its abstract norm mechanism, namely $1 + \sqrt{2}$. The examples in [35] show that this abstract mechanism alone cannot give 2. Related uniform-algebra abstractions encounter the same barrier [13]. Configuration-constant refinements give a strict improvement for every fixed numerical-range shape, but thin quadrilaterals make the relevant invariant approach its limiting value, so this method yields no smaller universal constant [30] and [37]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [31].

The missing information is algebraic. The companion in (5) is not an arbitrary operator: for the approximants constructed in Section 5, it belongs to the algebra generated by $f(B)^*$. Estimating it as an unrelated summand discards precisely this coupling.

1.2. *The new de-symmetrization mechanism.* Our argument retains the whole Cayley family, rather than the single identity (5). It has four steps.

(1) Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

(2) At an auxiliary stage, a vanishing perturbation arranges that $T = f(B)$ and $\operatorname{alg}(T) = \operatorname{alg}(B)$. The double-layer identity then gives the positive-real completion

$$(6) \quad H(w) - (I - wT)^{-1} \in \operatorname{alg}(T^*), \quad w \in \mathbb{D}.$$

(3) For the simple-spectrum approximants used at that stage, diagonalization turns the unknown term in (6) into a diagonal analytic correction. We sample the Herglotz kernel of H at half the conjugate eigenvalues and add one vector sample at the origin. The latter is chosen to cancel the diagonal correction exactly.

(4) The surviving positive-matrix inequality compares two weighted Gramians. Its difference is positive, and an eigenvector argument followed by a Stein identity yields $\|T\| \leq 2$.

A perturbation of f that vanishes in the limit enforces algebra generation on each simple-spectrum approximant. Successive limits in the function, the matrix, and the surrounding domain then yield the estimate for the originally arbitrary A and p ; no auxiliary condition remains in the conclusion.

2. Positive-real kernels

All matrix inequalities are in the Hermitian order.

A kernel $\mathcal{K} : \mathbb{D} \times \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ is positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$(7) \quad \sum_{i,j=0}^m \xi_i^* \mathcal{K}(w_i, w_j) \xi_j \geq 0.$$

We shall use the following standard matrix-valued form of the Herglotz theorem. Its kernel consequence is included to fix every normalization.

LEMMA 2.1 (Matrix Herglotz kernel). *Let $F : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$(8) \quad \mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}}$$

is a positive kernel.

Proof. The matrix Herglotz representation gives a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure M on $\mathbb{T}$, and a Hermitian matrix J_F , such that

$$(9) \quad F(w) = iJ_F + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} dM(\zeta).$$

For completeness, this follows by applying the scalar Herglotz theorem to $u^* F(\cdot) u$ for every $u \in \mathbb{C}^n$ and polarizing the resulting positive measures. Since the space is finite-dimensional, the polarized scalar measures are the entries of a unique finite matrix-valued measure; positivity follows from its quadratic forms.

For $|\zeta| = 1$, direct calculation gives

$$(10) \quad \frac{1}{1 - w\bar{z}} \left(\frac{\zeta + w}{\zeta - w} + \frac{\bar{\zeta} + \bar{z}}{\bar{\zeta} - \bar{z}} \right) = \frac{2}{(1 - w\bar{\zeta})(1 - \bar{z}\zeta)}.$$

The constant iJ_F cancels from (8). Substituting (9) and then summing as in (7) yields

$$2 \int_{\mathbb{T}} \left(\sum_i \frac{\xi_i}{1 - \bar{w}_i \zeta} \right)^* dM(\zeta) \left(\sum_j \frac{\xi_j}{1 - \bar{w}_j \zeta} \right) \geq 0.$$

This proves the assertion. $\square$

3. The positive-real completion principle

The next auxiliary theorem is the new core of the proof. It converts an algebraically constrained positive-real completion of a resolvent into the sharp norm bound. Section 4 constructs the required function at the algebra-generating stage, and Section 5 reaches that stage by perturbation before passing to arbitrary matrices and polynomials.

THEOREM 3.1 (Positive-real completion). *Let $T \in \mathbb{M}_n(\mathbb{C})$ have n distinct eigenvalues, all in $\mathbb{D}$. Let $H : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy*

$$(11) \quad H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}),$$

and

$$(12) \quad H(w) - (I - wT)^{-1} \in \operatorname{alg}(T^*) \quad (w \in \mathbb{D}).$$

Then $\|T\| \leq 2$.

Proof. Choose a diagonalization

$$(13) \quad T = S\Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad G = S^*S.$$

The eigenvalues λ_j are distinct and belong to $\overline{\mathbb{D}}$. Polynomial interpolation shows that

$$(14) \quad \operatorname{alg}(T^*) = \{(S^{-1})^*DS^* : D \text{ is diagonal}\}.$$

The inverse of the displayed linear parametrization is fixed and linear. Hence (12), applied to an analytic matrix function, gives an analytic diagonal function

$$\Theta(w) = \operatorname{diag}(\theta_1(w), \dots, \theta_n(w)), \quad \Theta(0) = 0,$$

such that, on setting $\widetilde{H}(w) = S^*H(w)S$,

$$(15) \quad \widetilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G.$$

Congruence by S preserves positive real parts. Consequently, Lemma 2.1 applies to $\widetilde{H}$.

Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$(16) \quad P_{ij} = \frac{G_{ij}}{1 - \overline{\lambda_i}\lambda_j/4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \overline{\lambda_i}\lambda_j/2}, \quad Y = Q - P.$$

Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$(17) \quad w_i = \frac{\overline{\lambda_i}}{2}, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n),$$

and add the point $w_0 = 0$ with vector

$$(18) \quad \xi_0 = v = -G^{-1}Pu.$$

We first verify that this sample cancels the entire unknown function Θ . The contribution of the second term in (15) to the kernel quadratic form is

$$(19) \quad 2 \operatorname{Re} \sum_{i=1}^n \overline{u_i} \theta_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \overline{w_j}} \right).$$

Because

$$1 - w_i \overline{w_j} = 1 - \overline{\lambda_i} \lambda_j / 4,$$

the parenthesis in (19) is $(Gv + Pu)_i$, which vanishes by (18).

It remains to calculate the contribution of $G(I - w\Lambda)^{-1}$. Between w_i and w_j its (i, j) entry is

$$(20) \quad \frac{2G_{ij}}{(1 - \overline{\lambda_i} \lambda_j / 2)(1 - \overline{\lambda_i} \lambda_j / 4)} = (4Q - 2P)_{ij}.$$

Between w_i and 0 the corresponding row is that of $G + Q$, and the block at $(0, 0)$ is $2G$. Kernel positivity and (18) therefore imply

$$(21) \quad \begin{aligned} 0 &\leq u^*(4Q - 2P)u + 2 \operatorname{Re}(u^*(G + Q)v) + 2v^*Gv \\ &= u^*(4Y - YG^{-1}P - PG^{-1}Y)u. \end{aligned}$$

As u was arbitrary,

$$(22) \quad 4Y - YG^{-1}P - PG^{-1}Y \succeq 0.$$

Balance the diagonalization by defining

$$(23) \quad \begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \widehat{P} &= G^{-1/2} P G^{-1/2}, \\ \widehat{Q} &= G^{-1/2} Q G^{-1/2}, & \widehat{Y} &= \widehat{Q} - \widehat{P}. \end{aligned}$$

Expanding the scalar kernels in (16) into geometric series gives

$$(24) \quad \widehat{P} = \sum_{k=0}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \widehat{Q} = \sum_{k=0}^{\infty} 2^{-k} \tilde{T}^{*k} \tilde{T}^k.$$

Indeed, $\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric weights give norm convergence. It follows that

$$(25) \quad \widehat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0.$$

Congruencing (22) by $G^{-1/2}$ yields

$$(26) \quad 4\widehat{Y} - \widehat{Y}\widehat{P} - \widehat{P}\widehat{Y} \succeq 0.$$

Let e be an eigenvector of the Hermitian matrix $\widehat{P}$, say $\widehat{P}e = \alpha e$. If $\alpha > 2$, then (26) gives

$$(27) \quad 0 \leq 2(2 - \alpha)e^* \widehat{Y} e.$$

By (25), this forces $e^*\widehat{Y}e = 0$. Positivity then gives $\widehat{Y}e = 0$; in particular, the $k = 1$ summand in (25) gives $\widetilde{T}e = 0$. Formula (24) now gives $\widehat{P}e = e$, contradicting $\alpha > 2$. The first series in (24) also begins with I , and hence

$$(28) \quad I \preceq \widehat{P} \preceq 2I.$$

Finally, the first Gramian satisfies the Stein identity

$$(29) \quad \widehat{P} = I + \frac{1}{4}\widetilde{T}^*\widehat{P}\widetilde{T}.$$

Using (28), we obtain

$$\widetilde{T}^*\widetilde{T} \preceq \widetilde{T}^*\widehat{P}\widetilde{T} = 4(\widehat{P} - I) \preceq 4I.$$

Thus $\|\widetilde{T}\| \leq 2$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U\widetilde{T}U^*,$$

where U is unitary. Therefore $\|T\| \leq 2$. $\square$

Remark 3.2. The factor $1/2$ in the sampling points (17) simultaneously produces the weights $1/2$ and $1/4$ in (24). Their positive difference and the Stein identity are what recover the sharp constant. The sample at the origin is not an estimate: it removes the unknown analytic correction exactly.

4. The double-layer completion

We now construct from the numerical range the function required by Theorem 3.1. The positive measure and the companion transform in this section are classical [24], [15], and [22]; the use of their full Cayley family to obtain (12) is the point needed here.

LEMMA 4.1 (Double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, and let $\Omega \subset \mathbb{C}$ be a bounded convex domain with continuously differentiable boundary and $W(B) \subset \Omega$. There is a unital positive map*

$$\Phi : C(\partial\Omega) \longrightarrow \mathbb{M}_n(\mathbb{C})$$

such that, whenever h is holomorphic on a neighborhood of $\overline{\Omega}$,

$$(30) \quad 2\Phi(h) = h(B) + g_h(B)^*,$$

where

$$(31) \quad g_h(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{h(\zeta)}}{\zeta - z} d\zeta, \quad z \in \Omega.$$

Proof. Orient $\partial\Omega$ counterclockwise. Let $\nu(\zeta)$ be the outward unit normal, let ds be arclength, and put

$$Z_\zeta = \zeta I - B, \quad \mathcal{R}_\zeta = Z_\zeta^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary measure

$$(32) \quad d\mu_B(\zeta) = \frac{1}{2\pi} \left(\nu(\zeta)\mathcal{R}_\zeta + \overline{\nu(\zeta)}\mathcal{R}_\zeta^* \right) ds.$$

It is positive. In fact, the supporting-line property of the convex set $\overline{\Omega}$ gives

$$(33) \quad \nu(\zeta)Z_\zeta^* + \overline{\nu(\zeta)}Z_\zeta = 2 \operatorname{Re}(\overline{\nu(\zeta)}Z_\zeta) \succeq 0.$$

Multiplication on the left by $\mathcal{R}_\zeta^*$ and on the right by $\mathcal{R}_\zeta$ turns the left side of (33) into the matrix in parentheses in (32).

Along the counterclockwise boundary, $d\zeta = i\nu(\zeta)ds$. The matrix Cauchy formula gives

$$(34) \quad \frac{1}{2\pi} \int_{\partial\Omega} \nu(\zeta)\mathcal{R}_\zeta ds = I.$$

Taking adjoints gives the same identity for the second layer. Hence

$$(35) \quad \int_{\partial\Omega} d\mu_B = 2I.$$

Consequently,

$$(36) \quad \Phi(\varphi) = \frac{1}{2} \int_{\partial\Omega} \varphi(\zeta) d\mu_B(\zeta)$$

is unital and positive. It is therefore star preserving.

For holomorphic h , the first layer of $2\Phi(h)$ is $h(B)$ by Cauchy's formula. The function g_h in (31) is holomorphic on Ω , so $g_h(B)$ is defined by the holomorphic functional calculus. To verify the relevant integral formula without assuming that g_h extends across $\partial\Omega$, choose a positively oriented contour $\Gamma \subset \Omega$ enclosing $\sigma(B)$. Cauchy's formula and Fubini's theorem give

$$(37) \quad \begin{aligned} g_h(B) &= \frac{1}{2\pi i} \int_{\Gamma} g_h(z)(zI - B)^{-1} dz \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \overline{h(\zeta)}(\zeta I - B)^{-1} d\zeta. \end{aligned}$$

Taking adjoints in (37) and using $d\bar{\zeta} = -i\overline{\nu(\zeta)}ds$ gives the second layer of $2\Phi(h)$. This proves (30). $\square$

The next construction is used only for auxiliary functions satisfying the displayed algebra identity. Section 5 enforces that identity by a vanishing perturbation and then passes to the original polynomial.

PROPOSITION 4.2 (Cayley completion for an auxiliary pair). *Let B and Ω be as in Lemma 4.1, and let f be holomorphic on a neighborhood of $\overline{\Omega}$ and*

satisfy

$$(38) \quad \max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1.$$

Put $T = f(B)$. If

$$(39) \quad \text{alg}(T) = \text{alg}(B),$$

then there is an analytic $H : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$(40) \quad H(0) = I, \quad \text{Re } H(w) \succeq 0 \quad (w \in \mathbb{D}),$$

and

$$(41) \quad H(w) - (I - wT)^{-1} \in \text{alg}(T^*) \quad (w \in \mathbb{D}).$$

Moreover, $\sigma(T) \subset \overline{\mathbb{D}}$.

Proof. Spectral inclusion follows from $\sigma(B) \subset \Omega$, spectral mapping, and (38). For $w \in \mathbb{D}$, define on the boundary

$$(42) \quad c_w(\zeta) = \frac{1 + wf(\zeta)}{1 - wf(\zeta)}$$

and put

$$(43) \quad H(w) = \Phi(c_w).$$

The map $w \mapsto c_w$ is analytic with values in $C(\partial\Omega)$, locally uniformly on $\mathbb{D}$. Hence H is analytic, and $H(0) = I$. Furthermore,

$$\text{Re } c_w(\zeta) = \frac{1 - |wf(\zeta)|^2}{|1 - wf(\zeta)|^2} \geq 0.$$

The positivity and star preservation of Φ give $\text{Re } H(w) = \Phi(\text{Re } c_w) \succeq 0$.

The uniformly convergent boundary expansion

$$(44) \quad c_w = 1 + 2 \sum_{m=1}^{\infty} w^m f^m$$

and (30) yield, with norm convergence,

$$(45) \quad H(w) = I + \sum_{m=1}^{\infty} w^m (T^m + g_{f^m}(B)^*).$$

Since the spectral radius of wT is less than 1, the Neumann series for $(I - wT)^{-1}$ converges in norm. Subtracting it from (45) gives

$$(46) \quad H(w) - (I - wT)^{-1} = \sum_{m=1}^{\infty} w^m g_{f^m}(B)^*.$$

Each g_{f^m} is holomorphic on a neighborhood of $\sigma(B)$. Finite-dimensional holomorphic functional calculus, equivalently Hermite interpolation followed by Cayley–Hamilton, implies $g_{f^m}(B) \in \text{alg}(B)$. By (39), every partial sum on

the right of (46) belongs to $\text{alg}(T^*)$. This algebra is finite-dimensional and therefore norm closed, which proves (41). $\square$

Combining the preceding results gives the sharp estimate at the auxiliary simple-spectrum, algebra-generating stage. The next section constructs this stage arbitrarily close to the original data and then passes to the limit.

COROLLARY 4.3 (Auxiliary sharp bound). *In Proposition 4.2, if $T = f(B)$ has distinct eigenvalues, then $\|f(B)\| \leq 2$.*

Proof. Apply Theorem 3.1 to the function H constructed in Proposition 4.2. $\square$

5. Passage to arbitrary matrices and proof of the main theorem

We now pass from the auxiliary stage to the arbitrary matrix and polynomial in Theorem 1.1. Two elementary approximation facts supply simple-spectrum matrices and smooth surrounding domains; the proof then enforces algebra generation by a vanishing perturbation.

LEMMA 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$(47) \quad W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}.$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries. Its nonvanishing set is therefore dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (47). $\square$

LEMMA 5.2 (Outer approximation). *If $K \subset \mathbb{C}$ is nonempty, compact, and convex, then there are bounded convex domains Ω_j with continuously differentiable boundaries such that*

$$(48) \quad K \subset \Omega_j, \quad d_H(\overline{\Omega_j}, K) \longrightarrow 0,$$

where d_H denotes the Hausdorff distance between compact sets.

Proof. For $\delta > 0$, let

$$\Omega_\delta = \{z \in \mathbb{C} : \text{dist}(z, K) < \delta\}.$$

This is a bounded convex domain and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$, so its Hausdorff distance from K is δ . The metric projection $\pi_K : \mathbb{C} \rightarrow K$ is unique and continuous. The squared distance to K is continuously differentiable off K , with gradient $2(z - \pi_K z)$. Its gradient does not vanish on the level

set $\text{dist}(z, K) = \delta$. The implicit-function theorem therefore shows that $\partial\Omega_\delta$ is continuously differentiable. Taking $\delta \downarrow 0$ proves the lemma. $\square$

We now prove Theorem 1.1.

Proof of Theorem 1.1. Let A and p be arbitrary as in Theorem 1.1. Choose a bounded convex domain Ω with continuously differentiable boundary such that

$$(49) \quad W(A) \subset \Omega.$$

We first prove

$$(50) \quad \|p(A)\| \leq 2 \max_{z \in \bar{\Omega}} |p(z)|.$$

Because $W(A)$ is compactly contained in Ω ,

$$\text{dist}(W(A), \mathbb{C} \setminus \Omega) > 0.$$

Lemma 5.1 and (47) therefore give a sequence of simple-spectrum matrices $B_k \rightarrow A$ with $W(B_k) \subset \Omega$. Since $p(B_k) \rightarrow p(A)$, it is enough to prove (50) for an arbitrary such matrix B .

Write the distinct eigenvalues of B as $\beta_1, \dots, \beta_n$ and set

$$(51) \quad m_\Omega = \max_{z \in \bar{\Omega}} |p(z)|.$$

If $m_\Omega = 0$, the polynomial vanishes on the open set Ω and is identically zero. In the remaining case $m_\Omega > 0$, put

$$(52) \quad f = \frac{p}{m_\Omega}, \quad R_\Omega = \max_{z \in \bar{\Omega}} |z|.$$

For $\eta \in \mathbb{C}$ define

$$(53) \quad f_\eta(z) = \frac{f(z) + \eta z}{1 + |\eta| R_\Omega}.$$

Then $\max_{z \in \bar{\Omega}} |f_\eta(z)| \leq 1$. For a pair $i \neq j$, the equality

$$f(\beta_i) + \eta \beta_i = f(\beta_j) + \eta \beta_j$$

excludes at most one value of η . There are thus only finitely many exceptional values for which the numbers $f_\eta(\beta_1), \dots, f_\eta(\beta_n)$ fail to be distinct. Choose nonexceptional $\eta_\ell \rightarrow 0$ and put

$$(54) \quad T_\ell = f_{\eta_\ell}(B).$$

Each T_ℓ has distinct eigenvalues. It also generates the same unital algebra as B . Indeed, $T_\ell \in \text{alg}(B)$, while Lagrange interpolation at the distinct points $f_{\eta_\ell}(\beta_i)$ supplies a polynomial q_ℓ satisfying

$$q_\ell(f_{\eta_\ell}(\beta_i)) = \beta_i \quad (1 \leq i \leq n).$$

Since B is diagonalizable, this implies $q_\ell(T_\ell) = B$. Consequently,

$$(55) \quad \text{alg}(T_\ell) = \text{alg}(B).$$

Corollary 4.3, applied to B and f_{η_ℓ} , gives

$$\|f_{\eta_\ell}(B)\| \leq 2.$$

Letting $\ell \rightarrow \infty$ yields $\|f(B)\| \leq 2$, or

$$(56) \quad \|p(B)\| \leq 2m_\Omega.$$

Letting the simple-spectrum matrices B_k tend to A proves (50).

Apply Lemma 5.2 to the compact convex set $K = W(A)$, obtaining Ω_j . All these closures eventually lie in one fixed compact neighborhood of K . Uniform continuity of p there and (48) give

$$(57) \quad \max_{z \in \Omega_j} |p(z)| \longrightarrow \max_{z \in W(A)} |p(z)|.$$

Apply (50) to each Ω_j and pass to the limit. This proves (3).

Thus the polynomial estimate holds for the originally arbitrary A and p . The simple-spectrum matrices, the algebra-generating perturbations, and the smooth surrounding domains occurred only inside approximating sequences and have all disappeared in the successive limits.

It remains to justify the spectral-set conclusion. Let r be rational with no pole on $W(A)$. Choose $\delta > 0$ such that r is holomorphic on the closed δ -neighborhood K_δ of $W(A)$. The compact convex set K_δ has connected complement, so polynomial Runge approximation gives polynomials p_j converging uniformly to r on K_δ . As $\sigma(A) \subset W(A) \subset \text{int } K_\delta$, the holomorphic functional calculus gives $p_j(A) \rightarrow r(A)$. Also

$$\max_{z \in W(A)} |p_j(z)| \longrightarrow \max_{z \in W(A)} |r(z)|.$$

Apply the polynomial estimate to p_j and pass to the limit to obtain (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

Theorem 1.1 immediately gives the usual holomorphic version on a neighborhood of the numerical range.

COROLLARY 6.1. *Let $A \in \mathbb{M}_n(\mathbb{C})$, and let f be holomorphic on a neighborhood of $W(A)$. Then*

$$(58) \quad \|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|.$$

Proof. Polynomial Runge approximation on a sufficiently small closed convex neighborhood of $W(A)$ gives the same limiting argument used at the end of the proof of Theorem 1.1. $\square$

The operator form follows by the standard finite-dimensional compression argument; compare [4].

COROLLARY 6.2 (Hilbert-space form). *Let $\mathcal{H}$ be a complex Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. For every $p \in \mathbb{C}[z]$,*

$$(59) \quad \|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. The assertion is immediate if $\mathcal{H} = \{0\}$ or $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x(A \upharpoonright_{\mathcal{M}_x}),$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$(60) \quad A_x^k x = A^k x \quad (0 \leq k \leq d),$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (59).

Put $K = \overline{W(A)}$. The standard inclusion $\sigma(A) \subset K$ shows that the rational functional calculus is defined for every rational function r without poles on K . Choose $\delta > 0$ so that r is holomorphic on the closed δ -neighborhood K_δ of K . This set is compact and convex, so polynomial Runge approximation gives polynomials p_j converging uniformly to r on K_δ . The holomorphic functional calculus gives $p_j(A) \rightarrow r(A)$, while

$$\sup_{z \in W(A)} |p_j(z)| \longrightarrow \sup_{z \in W(A)} |r(z)|.$$

Passing to the limit in (59) proves the rational estimate and hence the spectral-set assertion. $\square$

The conclusion controls errors in polynomial and rational approximations of matrix functions: if q approximates f uniformly on $W(A)$, then

$$(61) \quad \|f(A) - q(A)\| \leq 2 \max_{z \in W(A)} |f(z) - q(z)|.$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [12], [19], and [9] for the role of spectral sets in those settings.

7. Discussion and further directions

Theorem 1.1 proves exactly the original Crouzeix conjecture. Beyond this theorem, one may ask for a *uniform matrix-polynomial extension*: for every $m \geq 1$ and every matrix polynomial $\mathcal{F} = [f_{ij}] \in \mathbb{M}_m(\mathbb{C}[z])$, does one have

$$(62) \quad \|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\|, \quad \mathcal{F}(A) = [f_{ij}(A)].$$

Here $\|\mathcal{F}(A)\|$ is the operator norm on $(\mathbb{C}^n)^m$, whereas $\|\mathcal{F}(z)\|$ is the operator norm on $\mathbb{C}^m$. The case $m = 1$ is precisely the original conjecture proved above; uniformity in m is an additional operator-algebraic requirement. The foundational matrix-level disk estimate is due to Okubo and Ando [32]. Uniform matrix-level spectral-set bounds and their relation to numerical radius are studied in [23] and [27]; the constant $1 + \sqrt{2}$ at every matrix level follows from [22], while sharp ρ -contraction estimates are developed in [38]. The extension is known with constant 2 for weighted shift matrices [21]. The auxiliary simple-spectrum interpolation in Section 5 is eliminated by the limiting argument that proves Theorem 1.1. At the uniform matrix-polynomial level, however, the corresponding diagonal-algebra mechanism is not stable in the polynomial matrix size, making this a natural separate problem in operator-algebraic functional calculus.

The positive-real completion principle is independent of numerical ranges: it starts with a Carathéodory function whose resolvent defect lies in an adjoint-generated algebra. This separation suggests several concrete problems.

- (1) *Uniform matrix-polynomial extension.* Find a version of the diagonal-correction argument that is stable uniformly in the polynomial matrix size. Matrix-polynomial inputs lead to block coefficients that need not reduce to a single diagonal algebra, so the cancellation in (19) requires a genuinely operator-algebraic formulation.
- (2) *Infinite-dimensional completion theory.* A direct infinite-dimensional version of Theorem 3.1, based for example on spectral measures or Riesz bases, could reveal equality cases or uniform matrix-level functional-calculus structure. The main obstacle is to replace the finite diagonal correction and Lagrange generation (55) by a stable functional-calculus condition without relying on a diagonal spectral model. Related questions for unbounded numerical ranges have recently been studied in [18].
- (3) *Abstract uniform algebras.* The additional unitality structure isolated in [13] and [27] is close in spirit to (12). It is natural to ask whether positive-real completions yield the conjectured sharp constant in that abstract setting, or sharpen the spectral constants studied in [36] and [7].
- (4) *Other geometric functional calculi.* Double-layer methods apply to domains beyond numerical ranges and have recently been recast through

configuration constants and annular double-layer kernels [19], [37], [30], [28], and [2]. Elliptic domains also admit structured dilation models [1]. Whenever the companion transform lands in a controlled adjoint algebra, the kernel-sampling argument may replace a triangle-inequality estimate. This offers a route to sharp constants for annular, multiply connected, or shape-constrained spectral sets.

- (5) *Equality and stability.* With the notation of the proof of Theorem 3.1, if $\|\tilde{T}\| = 2$ and $e \in \mathbb{C}^n$ satisfies $\|e\|_2 = 1$ and $\|\tilde{T}e\|_2 = 2$, then equality throughout the last operator-inequality chain in the proof of Theorem 3.1 forces

$$\widehat{P}e = 2e, \quad \widehat{P}\tilde{T}e = \tilde{T}e.$$

Analyzing these vector saturation conditions may classify extremal pairs (A, p) and quantify stability near the nilpotent example (4). Such a classification must allow the nonuniqueness phenomena found in [29].

- (6) *Generalized numerical ranges.* Recent work proposes spectral-set problems for q -numerical ranges [33]. Their double-layer structure makes them a natural testing ground for positive-real completion, although no constant-2 analogue is asserted here.

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at half the conjugate eigenvalues and adding the origin sample that cancels the adjoint-algebra correction; this appears in the proof of Theorem 3.1, especially (17)–(19). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

References

- [1] J. Agler, Z. A. Lykova, and N. J. Young, *On the operators with numerical range in an ellipse*, J. Funct. Anal. **287** (2024), no. 8, Article 110556, 62 pp., <https://doi.org/10.1016/j.jfa.2024.110556>.
- [2] J. Agler, Z. A. Lykova, and N. J. Young, *Function theory on the annulus in the dp -norm*, Integral Equations Operator Theory **97** (2025), Article 30, <https://doi.org/10.1007/s00020-025-02814-w>.
- [3] C. Badea, M. Crouzeix, and B. Delyon, *Convex domains and K -spectral sets*, Math. Z. **252** (2006), no. 2, 345–365, <https://doi.org/10.1007/s00209-005-0857-y>.
- [4] K. Bickel, P. Gorkin, A. Greenbaum, T. Ransford, F. L. Schwenninger, and E. Wegert, *Crouzeix’s conjecture and related problems*, Comput. Methods Funct. Theory **20** (2020), no. 3–4, 701–728, <https://doi.org/10.1007/s40315-020-00350-9>.
- [5] K. Bickel, G. Corbett, A. Glenning, C. Guan, and M. Vollmayr-Lee, *Crouzeix’s conjecture, compressions of shifts, and classes of nilpotent matrices*, Concrete

- Operators **11** (2024), no. 1, Paper No. 20240004, <https://doi.org/10.1515/conop-2024-0004>.
- [6] K. Bickel and P. Gorkin, *Blaschke products, level sets, and Crouzeix's conjecture*, J. Anal. Math. **152** (2024), no. 1, 217–254, <https://doi.org/10.1007/s11854-023-0295-y>.
 - [7] H. Blazhko, D. Homza, F. L. Schwenninger, J. de Vries, and M. Wojtylak, *The algebraic numerical range as a spectral set in Banach algebras*, Canad. J. Math., published online 2025, <https://doi.org/10.4153/S0008414X25000124>.
 - [8] T. Caldwell, A. Greenbaum, and K. Li, *Some extensions of the Crouzeix–Palencia result*, SIAM J. Matrix Anal. Appl. **39** (2018), no. 2, 769–780, <https://doi.org/10.1137/17M1140832>.
 - [9] T. Chen, A. Greenbaum, and T. Trogdon, *GMRES, pseudospectra, and Crouzeix's conjecture for shifted and scaled Ginibre matrices*, Math. Comp. **94** (2025), no. 351, 241–261, <https://doi.org/10.1090/mcom/3963>.
 - [10] D. Choi, *A proof of Crouzeix's conjecture for a class of matrices*, Linear Algebra Appl. **438** (2013), no. 8, 3247–3257, <https://doi.org/10.1016/j.laa.2012.12.045>.
 - [11] A. Greenbaum and D. Choi, *Crouzeix's conjecture and perturbed Jordan blocks*, Linear Algebra Appl. **436** (2012), no. 7, 2342–2352, <https://doi.org/10.1016/j.laa.2011.12.005>.
 - [12] D. Choi and A. Greenbaum, *Roots of matrices in the study of GMRES convergence and Crouzeix's conjecture*, SIAM J. Matrix Anal. Appl. **36** (2015), no. 1, 289–301, <https://doi.org/10.1137/140961742>.
 - [13] R. Clouâtre, M. Ostermann, and T. Ransford, *An abstract approach to the Crouzeix conjecture*, J. Operator Theory **90** (2023), no. 1, 209–221, <https://doi.org/10.7900/jot.2021nov15.2364>.
 - [14] M. Crouzeix, *Bounds for analytical functions of matrices*, Integral Equations Operator Theory **48** (2004), no. 4, 461–477, <https://doi.org/10.1007/s00020-002-1188-6>.
 - [15] M. Crouzeix, *Numerical range and functional calculus in Hilbert space*, J. Funct. Anal. **244** (2007), no. 2, 668–690, <https://doi.org/10.1016/j.jfa.2006.10.013>.
 - [16] M. Crouzeix, *Spectral sets and 3×3 nilpotent matrices*, in C. Badea, D. Li, and V. Petkova (eds.), *Topics in Functional and Harmonic Analysis*, Theta Ser. Adv. Math., vol. 14, Theta, Bucharest, 2013, pp. 27–42.
 - [17] M. Crouzeix, *Some constants related to numerical ranges*, SIAM J. Matrix Anal. Appl. **37** (2016), no. 1, 420–442, <https://doi.org/10.1137/15M1020411>.
 - [18] M. Crouzeix, *Numerical ranges and spectral sets: the unbounded case*, preprint, 2025, arXiv:2509.19792, <https://arxiv.org/abs/2509.19792>.
 - [19] M. Crouzeix and A. Greenbaum, *Spectral sets: numerical range and beyond*, SIAM J. Matrix Anal. Appl. **40** (2019), no. 3, 1087–1101, <https://doi.org/10.1137/18M1198417>.
 - [20] M. Crouzeix, A. Greenbaum, and K. Li, *Numerical bounds on the Crouzeix ratio for a class of matrices*, Calcolo **61** (2024), no. 2, Paper No. 32, 18 pp., <https://doi.org/10.1007/s10092-024-00580-6>.

- [21] M. Crouzeix and A. Greenbaum, *A new proof that the numerical range is a complete 2-spectral set for weighted shift matrices*, preprint, 2025, arXiv:2508.12768, <https://arxiv.org/abs/2508.12768>.
- [22] M. Crouzeix and C. Palencia, *The numerical range is a $(1 + \sqrt{2})$ -spectral set*, SIAM J. Matrix Anal. Appl. **38** (2017), no. 2, 649–655, <https://doi.org/10.1137/17M1116672>.
- [23] K. R. Davidson, V. I. Paulsen, and H. J. Woerdeman, *Complete spectral sets and numerical range*, Proc. Amer. Math. Soc. **146** (2018), no. 3, 1189–1195, <https://doi.org/10.1090/proc/13801>.
- [24] B. Delyon and F. Delyon, *Generalization of von Neumann’s spectral sets and integral representation of operators*, Bull. Soc. Math. France **127** (1999), no. 1, 25–41, <https://doi.org/10.24033/bsmf.2340>.
- [25] C. Glader, M. Kurula, and M. Lindström, *Crouzeix’s conjecture holds for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue*, SIAM J. Matrix Anal. Appl. **39** (2018), no. 1, 346–364, <https://doi.org/10.1137/17M1110663>.
- [26] A. Greenbaum and M. L. Overton, *Numerical investigation of Crouzeix’s conjecture*, Linear Algebra Appl. **542** (2018), 225–245, <https://doi.org/10.1016/j.laa.2017.04.035>.
- [27] M. Hartz and J. E. McCarthy, *From Clouâtre–Ostermann–Ransford to Okubo–Ando*, preprint, 2026, arXiv:2606.02922, <https://arxiv.org/abs/2606.02922>.
- [28] M. T. Jury and G. Tsikalas, *Positivity conditions on the annulus via the double-layer potential kernel*, Studia Math. **278** (2024), no. 3, 233–265, <https://doi.org/10.4064/sm231023-2-8>.
- [29] K. Li, *On the uniqueness of functions that maximize the Crouzeix ratio*, Linear Algebra Appl. **599** (2020), 105–120, <https://doi.org/10.1016/j.laa.2020.03.045>.
- [30] B. Malman, J. Mashregi, R. O’Loughlin, and T. Ransford, *Double-layer potentials, configuration constants, and applications to numerical ranges*, Int. Math. Res. Not. IMRN **2025** (2025), no. 8, Paper No. rna084, 34 pp., <https://doi.org/10.1093/imrn/rna084>.
- [31] B. Malman, J. Mashregi, R. O’Loughlin, and T. Ransford, *On the Crouzeix ratio for $N \times N$ matrices*, preprint, 2024, arXiv:2409.14127, <https://arxiv.org/abs/2409.14127>.
- [32] K. Okubo and T. Ando, *Constants related to operators of class C_ρ* , Manuscripta Math. **16** (1975), no. 4, 385–394, <https://doi.org/10.1007/BF01323467>.
- [33] R. O’Loughlin and J. Rani, *q -numerical ranges and spectral sets*, preprint, 2026, arXiv:2603.15536, <https://arxiv.org/abs/2603.15536>.
- [34] R. O’Loughlin and J. Virtanen, *Crouzeix’s conjecture for classes of matrices*, Linear Algebra Appl. **697** (2024), 277–292, <https://doi.org/10.1016/j.laa.2023.12.008>.
- [35] T. Ransford and F. L. Schwenninger, *Remarks on the Crouzeix–Palencia proof that the numerical range is a $(1 + \sqrt{2})$ -spectral set*, SIAM J. Matrix Anal. Appl. **39** (2018), no. 1, 342–345, <https://doi.org/10.1137/17M1143757>.
- [36] F. L. Schwenninger and J. de Vries, *On abstract spectral constants*, in M. Ptak, H. J. Woerdeman, and M. Wojtylak (eds.), *Operator and Matrix Theory, Function*

- Spaces, and Applications*, Oper. Theory Adv. Appl., vol. 295, Birkhäuser/Springer, Cham, 2024, pp. 335–350, https://doi.org/10.1007/978-3-031-50613-0_15.
- [37] F. L. Schwenninger and J. de Vries, *The double-layer potential for spectral constants revisited*, Integral Equations Operator Theory **97** (2025), no. 2, Paper No. 13, 22 pp., <https://doi.org/10.1007/s00020-025-02800-2>.
- [38] F. L. Schwenninger and J. de Vries, *A sharp bound for the functional calculus of ρ -contractions*, Proc. Amer. Math. Soc. **153** (2025), no. 11, 4759–4768, <https://doi.org/10.1090/proc/17346>.

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